

AI's growing impact on Fraud Analysis and Fraud Detection

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****Course:**** BUS 751 – Python for Business Analytics

****Date:**** September 20, 2026

****Pathway:**** Research-Oriented Investigation

****Target output:****

1. Introduction & Motivation

It can be argued that maintaining a strong focus on financial fraud detection is critical for the sake of organization and consumer alike. Several setbacks and consequences

are visited upon organizations and consumers alike when it comes to light that a case of financial fraud has taken place (ACFE, 2024; Karajian & Ullah, 2022).

Whether it be hundreds of thousands in US dollars lost to fraudulent reporting (ACFE, 2024) or the psychological damage caused to individuals who fall victim to fraud

(Kassem, 2024), continued efforts to better utilize available data in fraud prevention and detection efforts are warranted (Theodorakopoulos et al., 2024). Research

on the uses of machine learning in fraud detection saw a substantial increase in 2022 with a sharp spike in publications between 2023 and 2024 (Chen et al., 2025).

****Study question:**** *Do the growing impact of AI in the business environment or stances on AI impact the fraud analysis process, and if so, how does that stance reflect on fraud analysis?*

2. Pathway & Target Contribution

- ****Pathway:**** Research-Oriented Investigation

- ****Contribution in one sentence:**** After this study, the reader should have a better understanding of how fraud detection research may change and evolve over the next few years.

3. Literature Review & Research Gap

It can be argued that maintaining a strong focus on financial fraud detection is critical for the sake of organization and consumer alike. Several setbacks and consequences are visited upon organizations and consumers alike when it comes to light that a case of financial fraud has taken place (ACFE, 2024; Karajian & Ullah, 2022). Whether it be hundreds of thousands in US dollars lost to fraudulent reporting (ACFE, 2024) or the psychological damage caused to individuals who fall victim to fraud (Kassem, 2024), continued efforts to better utilize available data in fraud prevention and detection efforts are warranted (Theodorakopoulos et al., 2024). Research on the uses of machine learning in fraud detection saw a substantial increase in 2022 with a sharp spike in publications between 2023 and 2024 (Chen et al., 2025). In addition to recent literature studying possible applications for AI in the fraud detection process as the modern landscape evolves (Belahouaoui & Alm, 2025; Van Belle et al, 2023), such changes present fraud perpetrators are with new ways to devise more complex schemes and force fraud detection as a field to evolve just as rapidly (Pourhabibi et al., 2020; Lokanan & Maddhesia, 2025). According to a systematic literature review of 57 articles (2019-2024) by Chen et al. (2025), the current field of financial fraud detection is undergoing a transition away from traditional machine learning methods in favor of deep learning approaches. This evolution in the modern landscape is driven by the necessity for tools that can more effectively manage the complexities of contemporary finance and process non-linear data structures.

Chen et al. also highlight four hurdles limiting the utilization of deep learning; Data Imbalance, Explainability, Non-Stationarity, & Privacy. The issue of data imbalance is one I have become all too familiar with during my first year in this program. With approximately 84% of the analyzed papers citing "class imbalance" (fraudulent vs. legitimate transactions) as a major barrier, Chen et al. recommend using AUC-PR over standard Accuracy metrics to develop more reliable evaluations as a counter towards non-stationarity (2025). Further issues hindering full utilization of deep learning include a scarcity of "Explainable AI" (XAI), the evolving fraud patterns creating a gap between what models can observe and what is actually going on, and growing privacy software causing computational complexity and reducing data access. Chen et al. close out their article by advising future research to foster more cooperation between academia and financial institutions to better translate theoretical models into real-world utilization and practice (2025).

Looking at benefits and advances of implementing machine learning into the fraud detection process, Sarna et al. discuss using edge AI to mitigate hurdles and obstacles which would normally limit what humans can accomplish without such tools (2025). Another strategy strongly worth considering is federal mediation of different financial institutions in training shared fraud models. Through this method, organizations can contribute to developing deep learning models without risking a leak of sensitive user data, thus, maintaining compliance with privacy laws (Sarna et al., 2025). Sarna et al. address the need for standardized data preprocessing and evaluation frameworks so as to make better comparisons across studies (2025). Sarna et al. also advocate for edge processing as a means mitigate digital traffic

congestion during times of large consumer use, such as holiday shopping (2025). Just as Chen et al. (2025; 2026) have, Sarna et al. also touch on the shift from machine learning to deep learning while calling for future research to focus on improving the transparency of the AI model (2025). However, proper implementation of AI tools stalled by several factors, including three addressed by Sarna et al., (2025); lack of interpretability, lack of global data sharing, and malicious AI models. Whether due to failures in communication or from AI tools being just as available for fraudulent intent as for fraud prevention, the AI tools currently available still leave much room for improvement in the fraud prevention effort. Sarna et al. conclude their article by advocating for a hybrid system in which AI models can expedite processes where possible while leaving a means for humans to audit the machine-made decisions and make informed decisions as whether to alter such choices (2025).

Chen et al. bring more to the literature through a systematic review of 108 peer-reviewed publications (2019–2024) on utilizing deep learning in the financial fraud detection process (2026). The literature touches on developments and challenges with implementing deep learning into fraud detection, including the implementation of hybrid systems as suggested by Sarna et al. (2025). However, the challenges brought up by recent literature present especially difficult challenges to address before AI tools can be used to their full potential. Until new models start integrating Explainable AI (XAI), they cannot be legally defended during regulatory audits. Outside of solidifying standards and regulations, the large resource requirements to get and maintain such systems causes them to be less accessible for smaller banks or other entities of similar size with limited assets (Chen et al., 2026). As for the challenge of catching fraud perpetrators, the noted rise in AI vs. AI warfare has escalated to a point where fraud perpetrators are using generative models in an effort to trick the banks' models into not flagging certain anomalous transactions or files (Chen et al., 2026). Lastly, Chen et al. also touch on new applications companies should consider to utilize deep learning to its fullest and to better prepare for how the economy and landscape will change, including use of language processing models to safe-guard against potential fraud at the executive level (2026).

In short, recent financial fraud detection literature has put a strong focus on understanding what AI tools can do in the fraud detection process and what can be done to maximize its effectiveness. With the world changing as quickly as it is, consideration of AI tools has become less of a fad to ignore in favor of critical human thinking and more of a powerful tool that people are becoming less able to ignore if they wish to keep up. Such a trend is well exemplified by the rise of adversarial AI and its efforts to undermine security models. Because of this, both fraud examiners and fraud perpetrators are now in a form of digital arms race to get a better grip of this new tool humanity has developed and trying to get ahead of the other side. Future research will likely soon stop asking research questions along the lines of whether AI tools can be of aid, if it has not already done so. Future research will more likely focus on attempts to refine existing models, or efforts to develop new ones. At least until data access gets to a point where the issue with data imbalance can be addressed in favor of more direct data analysis research.

4. Research Question(s) & Hypotheses

****Primary research question:****

****Hypothesis 1 (H1 – predictive value):****

- ****H0:****

- ****Falsification condition:****

****Hypothesis 2 (H2 – business age):****

- ****H0:****

- ****Falsification condition:****

****Operationalization of key constructs****

| Construct | Measure |

5. Data

Sources

| Source | Variables / fields | Frequency | Coverage | Accessibility |

|---|---|---|---|---|

6. Methods & Identification Strategy

****Estimators / models****

****Identification strategy.****

****Validation & robustness****

7. Analysis Plan & Expected Outcomes

****Pre-specified analyses****

****What supports vs. refutes each hypothesis****

Hypothesis	Supported if...	Refuted if...
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H1	---	---
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H2	---	---
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8. Contribution & Significance

****Contribution to research.****

- ****Value to Research:**** Whether and how to approve and structure a 7(a) loan (amount, term, collateral, processing method), and where SBDC advisors should focus pre-application coaching.

- ****Value to Business:**** At a chosen review threshold, the model identifies a flagged group of applicants for closer underwriting. Value is estimated as $\text{*expected losses avoided} = (\text{charge-offs caught in the flagged group} \times \text{average charge-off amount}) - (\text{cost of extra review} + \text{interest income forgone on good loans declined})$ *, using actual charge-off amounts from the FOIA data in the test years. The dashboard lets a user move the threshold and watch this trade-off change.

****Success metrics (independent of the direction of results)****

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9. Threats to Validity, Limitations & Fallback

****Internal validity****

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****External and construct validity****

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****Limitations & fallback****

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10. Technical Implementation & Reproducibility

- ****Databases:****

- ****Analysis:**** Python 3.14+; pandas, numpy, statsmodels, scikit-learn, (optional) xgboost, shap, lifelines (Fine–Gray / survival), matplotlib/plotly. Numbered scripts in `scripts/` ('01_download.py' → '05_evaluate.py'), fixed random seeds, `requirements.txt`, and results written to `outputs/`.

- ****Dashboard:**** View results through Streamlit app.

- ****Deployment:**** Streamlit Community Cloud deployed from the GitHub repo.

Timeline & milestones

| Week(s) | Milestone / activities | Course anchor |

|---|---|---|

| 1-4 | Introduction to Python and conceptualization of Project | A 1&2 Survey and Project Skeleton|

| 5-6 | Introduction to Project topic and Methods | A3 Project Proposal|

| 7-8 | Build Project Outcome; profile missingness; descriptive EDA by year, sector, state | A4: Data Integration|

| 9-12 | Fit benchmark, logistic, and boosted models; out-of-time validation; bootstrap CIs; robustness and placebo tests; SHAP | A5&6: Analysis and interpretation of results |

| 13 | Final review, test for reproducibility check from scratch, polish write-up, submit | A7&Final: Finalized Project and Presentation |

11. References

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